

Dustin Van Tate Testa | dvtt.net | github.com/dvtate | linkedin.com/in/dvtt | +14785088225 | Greater Chicago

SKILLS

C++, C (since 2011): POSIX, Qt, Boost (contributor), parallel programming, modern C++, sqlite, build systems, libcurl
TypeScript, JavaScript (since 2014): WebAssembly, express, WebGL, webpack, WebSockets, echarts, PM2, NPM, REST
Advanced Linux User: managing servers since 2016, software distribution & packaging, filesystems, systemd, remote access
Other Languages: Python, BASH, HTML, CSS, SQL, SVG, LaTeX, Haskell, POV-Ray, RPM SPEC, Spanish, Danish
Tools: git, nginx, SSH, kdenlive, Wings3D, SketchUp, spreadsheets, Tailscale, Vim, GDB, valgrind, CMake, VNC, ollama, rclone, qmake

RELEVANT WORK EXPERIENCE

Software Engineer, Information Security Corporation

January 2022-Present

- Joined codebase spanning decades of development with several hundred thousand lines of code
- Working with kernel modules, software plugins, desktop applications, automation scripts, etc.
- Software used in national security critical applications
- Skills Used: C++, C, Linux, Qt, BASH, Python, RHEL, Packaging & Build Systems, POSIX, PKI, LDAP, NFS, CIFS

Freelance Software Developer/Engineer/Consultant, Upwork

May 2021-December 2021

- Worked with many clients while in college. Received Upwork's "Rising Talent" and "Top Rated" badges
- Skills used: TypeScript, C, C++, Python, BASH, Haskell, AWS, SQL, Webpack, ORM, System Administration
- Achievements: Upwork's "Rising Talent" and "Top Rated" badges

Software Developer, CAEMate

June 2020-December 2020

- Helped refactor 3D editor for WeStatiX.com resulting in significant performance gains
- Skills: JavaScript, WebGL, WebAssembly, jQuery, C++, Linux, Docker, git, threejs, software engineering

Senior Software Engineer, StrattyX (YC W20)

April 2019 – December 2019

- Contacted by CEO with a name, logo and dream to democratize algorithmic trading
- Led development of company's software
- Designed and implemented a novel distributed language system to power the company's product
- Built out product at Harvard Innovation Lab; Pitched company at TechCrunch Disrupt 2019 where we were finalists
- Skills used: Startups, marketing, SWE, PM, AWS, SQL, Node.js, React Native, Python, REST, finance, NLP, etc.

IT Apprentice, Macon-Bibb Board of Education

August 2016 – May 2018

Summer Camp Councillor, Mercer University Creative Computer Camps

2016 - 2018

PROJECTS

FIY, fiy.to ; git.fiy.to/fiy/fiy

2024 – Active

- A federated app platform and protocol with several usable apps including GitHub alternative, vCard manager, chat, etc.
- Highly optimized, massively parallel codebase that leverages intimate knowledge of HTTP protocol and Linux

Postfix Haskell, github.com/dvtate/postfix-haskell

2020 – 2023

- A low-level, stack-based, functional language that compiles to WebAssembly, compiler written in TypeScript
- Moving, Tracing Garbage Collector Written by hand in WebAssembly

SCL, github.com/dvtate/SCL

2020

- A bytecode-compiled language with a comfortable syntax that's focused on async programming
- Custom Parser, Compiler, VM, Garbage Collector, Event Loop, etc. written in C++

Corki Discord Bot, corki.js.org

2017 - 2019

- A Discord chat bot with over 200k users, a language-based rule-engine, web portal, RSS forwarding, game integrations, etc.

Websites

- ResearchInEnglish.com (2023) A website that uses AI to write news articles about open-source research
- XSS.software (2022) A distributed, serverless cloud powered by browser tabs
- More: stonks.bodge.dev (2021), xTie.net (2020), corki.js.org (2019), robobibb.github.io (2017), old.dvtt.net (2014)

EDUCATION

Illinois Institute of Technology

Chicago, IL

B.S. Computer Science (GPA: 3.37)

August 2018-December 2021

- 2018 Deans List, Heald Scholar
- Programming Lead for NASA Robotic Mining Competition Team (2019-2020)
- Computer Science Teaching Assistant (2019)

Howard High School

Macon, GA

Foreign Language Pathway; GPA: 3.86 (top 5%); 11 AP classes

August 2014-May 2018

- Highest Average: AP Computer Science, AP World History, AP Chemistry
- Clubs: Robotics (programming lead); BETA, FBLA, News Team, etc.